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05/02/2024

Data Analysis Without Coding: Assignment 2

1. Background Information

I've decided to conduct an analysis on the top-rated movies and TV shows accessible across the respective streaming services: Amazon Prime, Netflix, Hulu and Disney+. The objective is to thoroughly evaluate their respective ratings and conclusively determine which streaming service offers the most value for subscription (If you can only subscribe to one). Different people have different interests and obligations, some like comedy and others drama, some have a lot of time on their hands, others a couple dozen minutes here and there. Hopefully by the end of this report, I'll have convinced you why this dashboard could be used by anyone to determine which platform is best suited for them

2. Explore & Clean the Data

I have chosen one main dataset containing the 4 platforms, the movies available on them, as well as their rotten tomato scores. I have separate datasets that include more movies from each service as well as tv shows, and lastly, one final dataset which includes IMDB ratings for some variety in the scoring. All datasets were taken from Kaggle.

Below I included a list of the tasks performed in order to clean and correct said data:

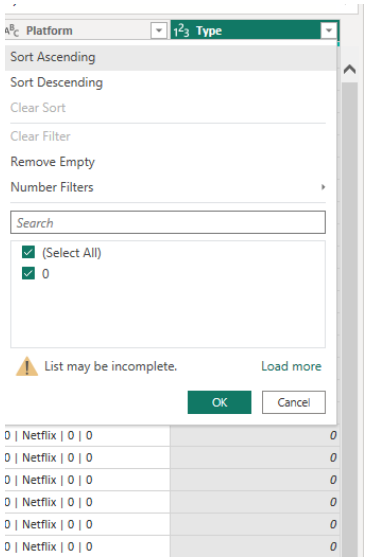
- The main dataset had a formatting issue where instead of one column telling us which platform a movie belongs to, it had 4 columns with Boolean values…

Netflix	Hulu	Prime Video	Disney+
1	0	0	0
1	0	0	0
1	0	0	0
1	0	0	0

  …this issue was corrected by changing the values of 1 to the name of the platform, the value 0 to a blank space, and then performing a merge so that they all appeared in the same column.
- There were certain columns that were deleted that either didn't have any relevance to the project or seemed liked errors:
- To work with the rotten tomato scores, a simple change was done whereby all /100 were erased and the type of the column set to numerical

Rotten Tomatoes
98/100

Rotten Tomatoes
98



- Certain movies/shows had multiple genres listed so in order to be able to better analyse based on this field, the first genre was taken as the primary while the others as secondary:

genres
animation, family, music, fantasy

genres.1	genres.2
animation	family, music, fantasy

- The Prime Video extended dataset included over 100 rows of incorrect data (eg. Year field coming in with a value of “warlocks”). It was decided that none of the affected movies would play a significant part in the analysis since the total number of items exceeded 9000 and so it was deemed safe to remove them.
- Next to add in the extra datasets, all headings were first normalized, a new column was added for the platform, at which point the “append query” did the rest of the work stacking all rows on top of each other. From there a simple "Merge Queries” Based on movie title is enough to have them all included.
- Entries with a null score for both IMDB and Rotten Tomatoes were deleted as they would not contribute to the comparison

3. Data Modelling & Transformation

- It was my intention to use images for each of the streaming platforms so as to more easily identify them. Therefore, it was necessary to create a new table (using DAX) that had the name of the service and the link as two separate columns.

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1 Images =
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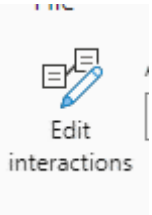
Production	
Prime Video	
Netflix	
Hulu	
Disney+	

The data category for these links was set to Image URL and so when used in my slicers, they showed up as intended.

- A list of extra columns added to assist data reporting are listed below:
  - Ranking: A categorical output based on the IMDB as well as the Rotten Tomato score.
  - Average Runtime: A normalised version of the original runtime column.
  - Type: Distinguishes between movie and tv show.
  - Custom: Used once to get rid of entries with neither an IMDB nor a Rotten Tomato rating.

- It was important in some cases to edit the interactions between visuals as certain slicers, for example, were put in place to serve only one graph.



4. Statistical Analysis

The dashboard is split into 3 sections: The main page, the Rotten Tomato related page and the IMDB related page. Pages can be scrolled through using a button at the top of each page.

The main page provides a general summary of the data, average scores, the spread of the data by genre and release year as well as age category. This is just to give a sense of what the data holds.

The two scoring pages provide all of the more detailed analytics. Here we have graphs that sort the tv shows by runtime and number of seasons. This would be important when determining which platform has the best short shows, for people with shorter attention spans (like myself) or longer ones for those who like to engage with the show.

We also have graphs that sort the media based on genre and age rating. This would be important for those who have particular preferences or perhaps have children they have to cater for.

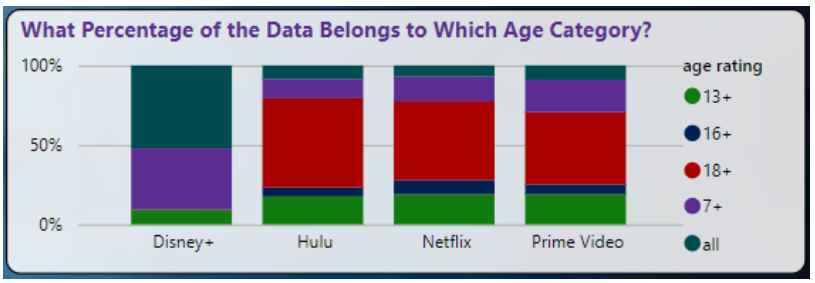
Lastly, we have a table that acts as a detailed source of information and gets updated based on what graph you are looking at, essentially giving you recommendations based on your selected slicers.

All pages come with various slicers and sliders that can help you determine what suits you. These range from platform selections at the top of the page, a slicer that switches between the different types of media, as well as sliders that help narrow down ratings.

5. Storytelling & Discussion

By studying the data and the various charts created using them, various insights were discovered:

- Hulu and Disney, although having less than half of the media that was recorded on either Prime or Netflix, came out with the highest average IMDB score and Rotten Tomato score respectively.
- The majority of Disney content is either suitable for all or 7+, meanwhile all other platforms have a similar spread that consists of mostly 18+ content, making Disney the obvious choice for those with children.



- As far as tv shows are concerned, IMDB ratings tell the best story, with Hulu having the best rated short tv shows. Meanwhile Disney+ leads the way for longer content
- In terms of number of seasons, Hulu is the choice for someone who enjoys longevity in a show, with multiple entries at 30+ seasons as well as boasting the show with the highest season count, “Coronation Street”, at 63. On the other end, it gets a bit more muddled as both Prime and Netflix fight for the top spot, both boasting multiple entries above a 9 on the IMDB scale.

- More generally, by genre on the IMDB side, Disney holds the lead in most categories, including crime interestingly enough, except for comedy and documentation which fall to Netflix. By Rotten Tomato's standards, things shift a bit more towards Hulu and at times Netflix.
- By age rating, Hulu and Disney+ once again fight it out for the top spot in most cases with Prime once again falling short due to its high abundance of bad media.

6. Conclusion

It might sound like this would be an easy choice to make, but the great thing about the dataset is that there are so many slicers to pick from that, depending on your preference, you could come out with completely different results than me. Take Prime for example, it scored horribly in comparison to the other 3 platforms, but this was when looking at all ratings. When we switch this to only consider the top ratings, both on the main page and each individual ratings page, the story changes as it becomes one of the better services in terms of quality, even the best in some cases. Take the following images as an example:

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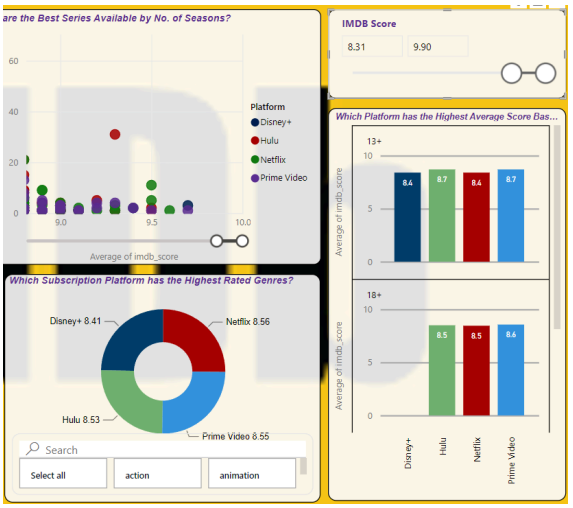

Before, prime was undoubtedly the worst off in terms of ratings whereas when we change the scale to only consider media that is rated 8.31 and above, we suddenly see it take the top spot in multiple places.

In conclusion, it depends on your perspective. Certain choices are easy, if you have kids, choose Disney or if you want to watch shows with 30+ seasons, choose Hulu. Other choices depend on preference. If you want to be able to find a good show or movie quickly without wasting time with bad ones, Disney or Hulu would be your choice. If you don't mind sifting through the junk, both Prime Video and Netflix offer some absolute gems in both the big and small screen departments, Prime in particular boasts the highest amount of media you can consume, according to the data. As for those with more particular tastes, a simple flick through some sliders will reveal exactly what you're looking for.